

1.3b

The neuron & neural firing Substance Use Disorders & Psychoactive drugs

What are substance use disorders?

Many can use Psychoactive Drugs, chemicals that alter the brain, in moderation.

Sometimes, drug use crosses the line between moderation & substance use disorder.

A person may get diagnosed w a SUD if drug use continues despite significant life disruptions

Resulting brain changes may persist after quitting use.
leading to strong cravings when memories of drug triggered

Indicators of SUD

Diminished Control

1. Uses more substance, or for longer than intended
2. Tries unsuccessfully to regulate use of substance
3. Spends much time acquiring, using or recovering from effects of substance
4. Craves the substance

Diminished Social Functioning

5. Use disrupts commitments at work, school, or home
6. Continues use despite social problems
7. Causes reduced social, recreational, and work activities

Hazardous use

8. Continues use despite hazards
9. Continues use despite worsening physical or psychological problems

Drug Action

10. experiences tolerance
11. experiences with withdrawal when attempting to end use

A drug's overall effect depends not only on biological effects, but also the user's expectations which vary socially & culturally.

Continued use of alcohol & some other drugs (but not weed) lead to tolerance :: brain chemistry adapts to offset drug effects

To experience the same effect, users need larger & larger doses.

Addiction prompts users to crave the drug, continue use despite adverse consequences, & struggle trying to withdraw

4% of everyone have an alcohol use disorder

After using, lifetime odds of getting hooked.

Marijuana	- 9%
Cocaine	- 21%
Alcohol	- 23%
Tobacco	- 68%

Most who successfully quit smoking, quit the habit on their own

Psychologists try to avoid using "addiction" to label other excessive behaviors, ex. eating, work, sex, capitalism

But some behaviors can become compulsive & dysfunctional similar to problematic drug use.

Behavior addictions include gambling disorder & internet gaming disorder.

There are 3 major categories of psychoactive drugs
Depressants, Stimulant, Hallucinogen

What are Depressants?

Depressants are drugs like alcohol, barbituates (tranquilizers) and opioids that calm neural activity & slow body functions

Alcohol

Alcohol is always a depressant & amounts

Small amounts may enliven, but do so by acting as a disinhibitor.

Alcohol slows brain activity that controls judgement and inhibitions - causing 3mil ^{deaths}/_{year}

Alc is an equal-opportunity drug

It increases (disinhibits) helpful tendencies, ex. social ↑

It increases harmful tendencies, drunk rapists are more likely to rape

One uni campus showed that 80% of male rapists 70% of female victims had been drinking.

Drinking increases desire for casual sex & perception of attractiveness

The urges you feel sober, you will be more likely to act on drunk

Alcohol use disorder contributes to >200 diseases & can shrink the brain, contribute to premature death.

Girls & young women, (who have less of a stomach enzyme that digests alc) can become addicted to alc more easily than boys & young men, and are at more risk for lung, brain & liver damage at lower consumption. 2001-2007, Canadian's women risk for alc-related death increased 5x the rate for men.

Alc slows SyNS activity.

larger doses → slurred speech, skilled performance to deteriorate.

Alc is a potent sedative, esp. alongside sleep deprivation.

When drunk, people aren't aware of how drunk they are.

heavy drinking can suppress the gag reflex. vomiting releases toxins, so this can be deadly causing people to poison themselves.

Alc can disrupt memory formation.

heavy drinking can have long term effects on the brain & cognition.

heavy drinking may cause blackouts $\therefore$ drinkers continue to interact but are unable to later recall people they met or things they did.
Partially because alc suppresses REM sleep.

Compared to a placebo, those who consumed alc were 2x likely to be mind-wandering, less likely to notice zoning out.

Expectations influence behavior

Assuming alc will lift spirits, adolescents may drink when upset & alone. Solitary drinking increases chance of developing a substance use disorder.

Believing or consuming alc can make u act like it

researchers gave uni men alc, or a drink the men thought was alc. After watching an erotic video, men who thought they had drunk were more likely to guiltlessly report having strong sexual fantasies.

Barbituates

like alc, barbituate drugs, which are tranquilizers depress nervous system activity.

Barbituates like Nembutal, Seconal & Amytal are sometimes prescribed to induce sleep or reduce anxiety.

In large doses, they can impair memory and judgement.

If combined with alc, the total depressive effect can be lethal.

Opioids.

Opioids, Opium & its derivatives, also depress neural functioning

Include Heroin, & methadone, its medically prescribed synthetic substitute.

They also include pain-relief narcotics like codeine, Oxycotin, Vicodin, morphine, & morphine's dangerous synthetic counterpart: fentanyl

Blossful pleasure replaces pain & anxiety.
Pupils constrict, breathing slows, lethargy sets in

Brain will eventually stop making endorphins, its own natural opioid, after repeated use.

If opioid is withdrawn, brain will lack normal level of these painkilling neurotransmitters.

between 2013-2016, opioid overdose deaths increased 10x to 43k, because the US is run by pharma companies & war mongers.

What are stimulants?

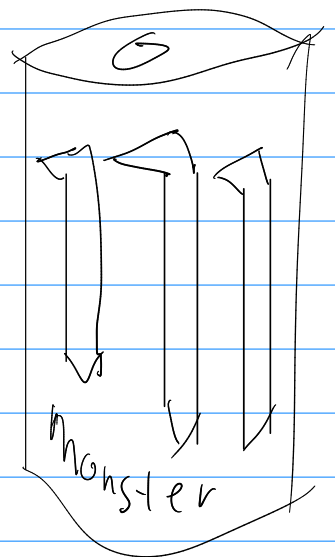
Stimulants excite neural activity & speed up body functions

Pupils dilate, heart rate increase, breathing rates increase, blood sugar levels rise, reducing appetite.

Energy & self confidence rise.

Personal connection.

I use stimulants a lot. Caffeine is a big part, but also my ADHD meds. Of all the different ADHD meds I've used, the only one that was not a stimulant was BY FAR, my worst experience with meds



Stimulants include caffeine, nicotine, cocaine, amphetamines, methamphetamine ('speed'), Ecstasy.

Nicotine

Highly addictive.

250 million packs per day consumed

teen-to-adult smoker has 50% chance of dying from it

Directly kills 7 million yearly, 1.2 million from second hand smoke.

Tobacco products: Cigarettes, cigars, chewing tobacco, pipe tobacco, snuff, e-cigarettes.

Vaping had rapid increase, fastest drug use increase on record.

High School students vaped 2.7 times rate of cigarettes

UK study, non-smoking teens who tried vaping 4x as likely to move to cigarettes

Fruity vape flavors, and other teen-focused marketing, target kids

Obviously Intentional.

these corporations don't care about dying kids

Smoker life expectancy at least 10 years shorter

Tobacco products are as powerfully and quickly addictive as heroin & cocaine.

Attempts to quit, even in first weeks, often fail

A rush of nicotine signals the CNS to release a flood of neurotransmitters.

Epinephrine & norepinephrine diminish appetite and boost alertness & mental efficiency.

Dopamine & opioids will temporarily calm anxiety & reduce pain sensitivity.

Major Depressive Disorder makes quitting harder

Cigarette smoking is the leading cause of preventable death in US, annually killing 480k.

$\frac{3}{4}$ smokers want to quit, annually $\frac{1}{7}$ do.

Worldwide smoking rate, 25% in men, 5% in women, down 30% since 1990

US rate: 45% in 1955 $\rightarrow$ 15% in 2014

Acute craving and withdrawal symptoms slowly dissipate over 6 months.

After quitting for 1 yr, only 10% will relapse

Cocaine

Cocaine is a powerfully addictive stimulant from the coca plant.

Coca-Cola originally had it.

Coke is snorted, injected or smoked

Sometimes smoked as Crack cocaine, faster crystallized form that is briefer but more intense, followed by a more intense crash.

Coke enters bloodstream quickly, producing rush of euphoria that depletes the brain's supply of dopamine, serotonin, norepinephrine.

Within 1 hour, crash of agitated depression follows.

After few hours craving wanes, but will return in a few days.

Coke binds to sites that reabsorb neurotransmitters, blocks reuptake

Coke may heighten reactions

Coke may lead to emotional disturbances, suspiciousness, convulsions, cardiac arrest or respiratory failure.

Placebo cocaine often gave coke-like experience

1% of US grade 12s & 6% of 18-24 yr olds in UK reported trying coke in past year

Methamphetamine

Amphetamines stimulate neural activity.

Body functions speed up, energy ↑, mood ↑

Methamphetamine is chemically similar to amphetamine but stronger

Meth triggers dopamine release. Stimulates brain cells that enhance energy & mood.

~8 hours of heightened energy & euphoria.

After effects may include irritability, insomnia, hypertension, seizures, social isolation, depression, occasional violent outbursts

meth reduces baseline dopamine over time

Ecstasy

Ecstasy is the street name for MDMA:
methylenedioxymethamphetamine. Powder = molly.

Triggers dopamine, but main: release & block
reuptake of serotonin

Effects felt 0.5 hour after taking pill.

3-4 hours duration; high energy, emotional elevation,
connectedness

Ecstasy is dehydrating
can damage serotonin-producing neurons
suppresses immune system
impairs memory
slows thought
disrupts sleep

What are Hallucinogens?

Hallucinogens distort perceptions & evoke sensory images in the absence of sensory input.

Some, ex. LSD, MDMA are synthetic

Others, ex. psilocybin, ayahuasca & weed are natural.

Brain hallucinates basically the same whether from drugs, loss of O_2 or extreme sensory deprivation

Experience typically begins with simple geometric forms

Then more meaningful images. (may be superimposed on a tunnel or replay of past emotional experiences)

Brain scans of LSD trippers showed hypersensitivity in the visual cortex, which strongly connected to emotional centers

At peak, people feel separated from their body, lose sense of self.

These are very similar to the near-death experience: altered state of consciousness reported by 10-23% of cardiac arrest survivors

Many describe visions of tunnels, bright lights, old memories

These can "enhance spirituality and promote feelings of personal growth"

LSD

emotions during an LSD ('acid') trip range from euphoria to detachment to panic.

Marijuana

weed leaves & flowers contain THC: delta-9-tetrahydrocannabinol.

Whether inhaled or consumed, THC produces a mix of effects.

The THC in a joint may induce psychiatric symptoms ex hallucinations, delusions, & anxiety.

Synthetic cannabinoids mimic THC.

weed amplifies sensitivity to colours, sounds, tastes & smells.

Like alc, it relaxes, disinhibits, may be euphoric

alc & weed impair motor coordination, perceptual skills, and reaction time necessary for safely operating a vehicle.

weed tolerance

body eliminates alc within hours
THC + byproducts linger for > week, repeated use accumulates in tissues

weed:

- alleviates: chronic pain, chemo related nausea, multiple sclerosis → muscle soreness
- short term sleep improvements
- doesn't increase risk of tobacco related diseases like lung cancer
- predicts: increased risk of traffic accidents, chronic bronchitis, psychosis, SAD, suicidal thoughts
- likely contributes to impaired attention, learning memory, bad academics

Young weed users 'neuro toxic'
predicted long term cognitive impairment

MCA: 1.d 2.b 3.c 4.c 5.c